Object Detection Algorithms: A Comparison

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2022 IEEE 4th International Conference on Civil Aviation Safety and Information Technology (ICCASIT) | 978-1-6654-6766-7/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICCASIT55263.2022.9986730

***Abstract*—Object detection, whose main task is to detect objects in a picture to determine the type, location, and scene to which they belong, has become one of the most central problems in computer vision. Object detection is currently broadly divided into two implementations, one-stage detector will get the prediction results directly from the image, two-stage detector is also known as Region-based, because they will regionally generate the image and then submit it to the convolutional neural network. These two approaches have their own advantages and disadvantages, and this paper will study their characteristics and summarize them with examples. This paper summarizes a total of five outstanding works on these two approaches, including YOLO, ConerNet, R-CNN, SPPNet, and Auto-FPN. The summary of these five efforts may help others better understand the field.**

Keywords—Object detection, one-stage detector, two-stage detector, Region proposal network

I. Introduction

The task of object detection is to find out all the targets of interest in the image and label the location and category of the objects, which is one of the core problems in the field of computer vision, but in practical applications, the shape, size, pose, color and other conditions of the objects vary greatly, and there may be lighting, occlusion, and other factors, so target detection is also one of the most challenging problems in the field of computer vision.

The deep convolutional neural network, alexnet [1], proposed by Krizhevsky et al. achieved record-breaking image classification accuracy in large-scale visual recognition challenge (ILSVRC) in 2015. Since then, most of the research focus in computer vision has been on specific deep learning methods, and indeed the area of generic target detection. With years of research, the field of object detection has made great progress. In practical applications, the selection needs to be decided between two possible object detection networks: one-stage networks and two-stage networks.

One-stage networks. In one-stage network, the CNN uses anchor boxes to make network predictions for regions of the entire image and decodes the predictions to generate the final bounding box of the object. Comparing with two-stage networks, single-stage networks would show their strength in speed, but they would be less competitive in accuracy, especially for scenes with small objects.

Two-stage networks. The initial stage of the two-stage network identifies region proposals, or a subset of images that may contain an object. The second stage classifies the objects in the region proposal. Two-stage networks can achieve very accurate object detection results; however, they are typically slower than one-stage networks.

There has been a lot of work focusing on object detection that has used both frameworks (single-stage and two-stage). In this paper, some of best of they will be summarized and investigated. To be specific, one-stage networks we have studied YOLO [2], ConerNet [3]. two-stage networks we have studied R-CNN [4], SPPNet [5], Auto-FPN.

Among studies used single-stage networks, there are some differences in the structure and implementation of YOLO and ConerNet. YOLO is inspired by the GoogLeNet[3] model for image classification. The detection network has 24 convolutional layers followed by 2 fully connected layers. Instead of the inception modules used by GoogLeNet, 1x1 reduction layers followed by 3x3 convolutional layers was used. CornerNet uses a single convolutional neural network to detect the bounding boxes of an object as the upper left and lower right corners. By detecting the top-left and top-right corners of objects as a pair of key points, there is no need to design a set of anchor boxes. In addition, ConerNet introduces the corner ensemble, a novel ensemble layer that helps the network to better localize corners. In the final experimental results, it was found that the ConerNet has better mean average precision, moreover accuracy rate it achieves is even comparable to two-stage networks, but the accelerated version of YOLO, fast YOLO is the fastest running object detector.

When it comes to studies chose two-stage networks, they are believed to have higher accuracy. R-CNN can be a good solution to the problem that it is difficult to pinpoint within a sliding window paradigm in a CNN that may have multiple convolutional layers, and can address the problem of data scarcity by performing supervised training on a large dataset followed by domain-specific fine-tuning on a small dataset. SPPNet adds an SPP layer to the last convolutional layer. The SPP layer will read information warping and clipping at the beginning. Auto-FPN will contain two modules, Auto-fusion and Auto-head. In Auto-fusion, there are SSD, FPN, PANet. Auto-fusion's search space can cover SSD, FPN and PANet, and it tries to seek a better fusion process through automatic neural architecture search. The Auto-head module is constructed by different auto-searched cells in order to perform a better classification and bounding box prediction. The results of the test show that they all have their advantages and disadvantages, R-CNN has the highest accuracy but requires the longest training time. The accuracy of SPPNet and R-CNN are basically comparable, but the running time of SPPNet is much shorter than that of R-CNN. However, these methods suffer low computation speed.

In this paper, we will examine and summarize their processes and results to get a clear view of the recent developments in object detection, including several of the most successful algorithms that have made breakthrough improvements and optimizations based on previous research and have shown their advantages in real-world evaluation.

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II. One-stage Detector

A. YOLO

Previous work on object detection has improved the accuracy of object detection by altering the way classifiers are used to implement it. As part of its novel approach to object recognition, YOLO frames object detection as a regression problem, which enables spatial separation of bounding boxes and association of class probabilities. In a single evaluation, a neural network's bounding box and class probabilities are predicted directly from the complete image. Due to the fact that the entire detection pipeline is composed of a single neural network, it can be optimized for detection performance in a single step.

As shown in Figure 1. YOLO is trained directly on an entire image, which enables not only simultaneous processing of numerous bounding boxes, but also direct optimization of detection performance.

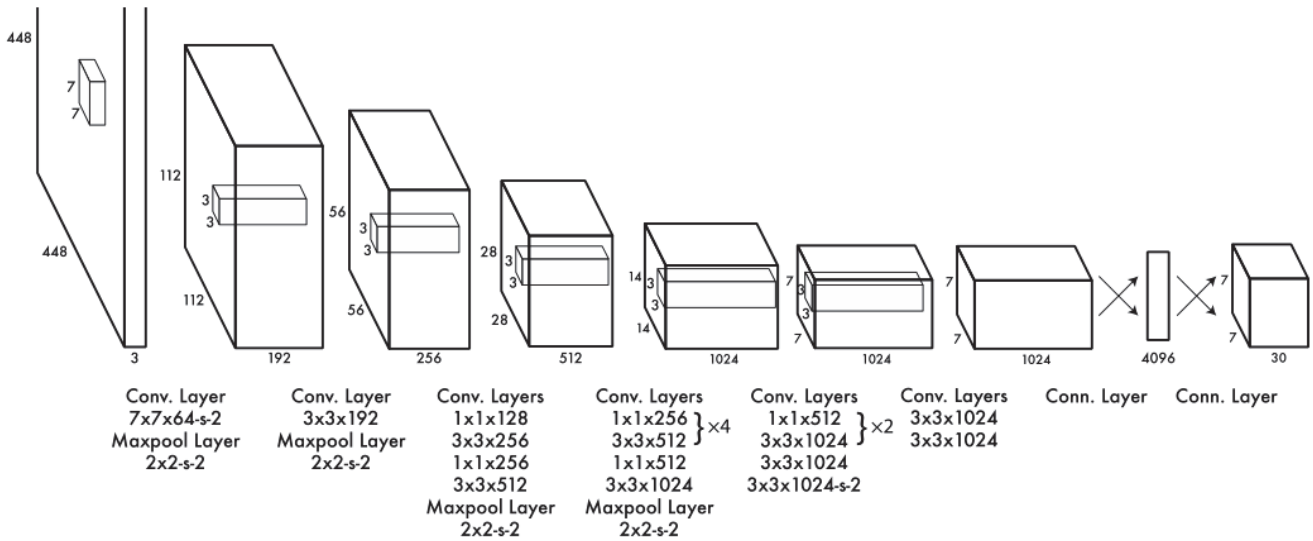

Fig.1 Overall architecture of YOLO.

The network structure of YOLO is inspired by the GoogLeNet image classification model. Yolo's detection network is comprised of 24 convolutional layers and two fully linked layers. YOLO updates GoogLeNet's receiver module with a basic 1x1 reduction layer and 3x3 convolutional layers. The initial few layers' feature space is reduced by the alternating 1x1 convolutional layers. In the ImageNet classification challenge, we pre-trained the convolutional layers at half the resolution (224x224 input picture) and then recognized them at twice the resolution..

PASCAL is a frequently used dataset for object recognition, and Fast YOLO is the fastest algorithm trained on it. The accuracy of mAP for Fast YOLO is more than double that of prior real-time detection efforts, reaching 52.7 %. YOLO increases mAP to 63.4 % while preserving real-time performance, as shown in Table I.

TABLE I PERFORMANCE ON PASCAL DATASET.

Real-Time Detectors	Train	mAP	FPS
Fast YOLO	2007+2012	52.7	155
YOLO	2007+2012	63.4	45
Less Than Real-Time			
R-CNN Minus R	2007	53.5	6
Fast R-CNN	2007+2012	70.0	0.5
Faster R-CNN VGG-16	2007+2012	73.2	7
Faster R-CNN ZF	2007+2012	62.1	18
YOLO VGG-16	2007+2012	66.4	21

Above all, YOLO will offer three primary benefits. To begin, YOLO is lightning fast. YOLO does not require a sophisticated pipeline because it approaches object detection as a regression problem. YOLO's basic version runs at 45 frames per second, but the fast version, Fast YOLO, runs at more than 150 frames per second, achieving double the mAP of competing real-time detectors. Second, when making predictions, YOLO performs global inference on the image, which means that it acts in such a way that it may encode contextual information about the category in addition to its look. YOLO has fewer than half the number of contextual errors as the rapid R-CNN. Thirdly, YOLO acquires representations of objects that are generalizable. This means that YOLO outperforms algorithms such as DPM and R-CNN when tested on artwork following picture training.

B. Cornernet

Convolutional neural network based work has been extremely successful in the field of object detection, with the most successful technique occurring when anchor boxes are shared. Anchor boxes are densely placed on the image during object detection, and then the anchor boxes are computed and the final box prediction is formed following regression computation. However, anchor boxes are not without their flaws. To begin, we typically require a large set of anchor boxes to guarantee that each anchor box overlaps adequately with the ground truth box, but this results in a huge number of anchor boxes being required to determine this, which slows down the training process. Second, anchor boxes introduce a plethora of hyperparameters and design options. This includes the number, size, and aspect ratio of the boxes, which are formed ad hoc using more computationally sophisticated heuristic algorithms. CornerNet, the new single-stage object identification approach, eliminates anchor boxes to address these concerns.

CornerNet is a novel approach for detecting objects. Yolo detects an object's bounding box as its upper left and bottom right corners using a single convolutional neural network. Prior to CornerNet, replacing objects with pairs of keypoints

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for detection was a standard way for replacing anchored boxes in single-stage detection experiments. Additionally, to improve corner localization, YOLO employs a new ensemble layer called the corner pooling layer, as shown in Figure 2.

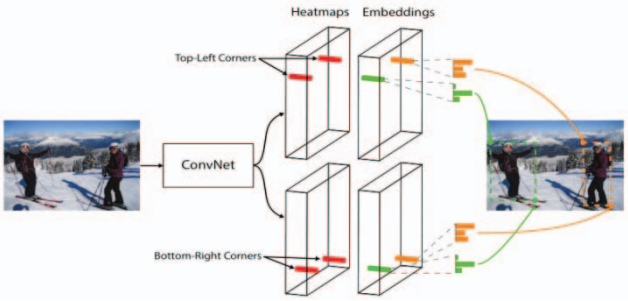

Fig.2 Network introduction of CornerNet.

CornerNet's object detection procedure begins by identifying the object's top-left and bottom-right corners and combining them into a pair of bounding box corners that are then sent to the convolutional network. The convolutional network generates a heat map of all the top-left and bottom-right corners and determines the embedding vector for each corner observed.

CornerNet incorporates corner sets as well. Often, the corner that defines a box is specified as being external to the object. In this situation, the location of a corner cannot be determined only on the basis of local data. Rather than that, to establish whether a pixel point has an upper-left corner, we must look to the right at the object's uppermost horizontal border and down at the object's leftmost vertical boundary..

Law et al. used the MS COCO dataset to evaluate CornerNet's performance and compare it to other state-of-the-art detectors. CornerNet produced an AP of 42.2 percent in the experiments, outperforming all previous single-stage detectors and even being equivalent to two-stage detectors in multi-scale evaluations.

III. Two-stage Detector

A. RCNN

PASCAL VOC is a well-known dataset for object detection testing that has increasingly stabilized in recent years. The complex ensemble system, which combines many low-level picture attributes with a high-level background, is one of the best performers. Thus, the research on R-CNN is an attempt to develop a simple and scalable detection method that outperforms the previous best findings in terms of average accuracy. The R-CNN has two objectives when implementing this algorithm. (1) When labeled training data is scarce, considerable performance increases can be obtained by applying supervised pre-training followed by domain-specific fine-tuning on auxiliary tasks.

R-CNN is the combination of CNN and region proposals module. High-level units in a CNN with numerous convolutional layers have extremely broad perceptual fields and spans in the input image, making exact localization challenging in the sliding window paradigm. At the time of testing, the R-CNN approach was to generate region proposals for around 2000 classes in the input image, extract a fixed-length feature vector from each proposal using the CNN, and then classify each region using a class-specific linear SVM. To generate a fixed-size CNN input from each region recommendation, R-CNN uses affine image warping, which ignores the image's form.

The second obstacle to overcome while performing experiments is data scarcity; the amount of data available at the moment is insufficient to train a big CNN, and the conventional technique is to employ unsupervised pre-training followed by supervised fine-tuning. On the other hand, Girshick et al. employed supervised training on a large dataset followed by domain-specific fine-tuning on a small dataset. The final results demonstrate that their strategy has the potential to greatly enhance average accuracy.

To test the effect of training on the PASCAL VOC dataset, Girshick et al. compare their results with the UVA system developed by Uijlings et al. They built a four-level spatial pyramid and used a multi-feature, nonlinear kernel SVM method for classification. Compared with UVA, the method adopted by the author has better mean average precision (53.7% vs 35.1%), and the speed is also faster.

To test the effect of training on ILSVRC2013, Girshick et al. used the same system hyperparameters that we used for PASCAL VOC. They compared the results of R-CNN with the works that participated in the ILSVRC2013 competition and the results of Overfeat (the second-best detector), and the mean average precision of R-CNN was nearly 7% higher than that of Overfeat's 24.3%, Although slower.

R-CNN implements a simple and scalable object detection technique that outperforms PASCAL VOC 2012 by 30%. It has two primary benefits: first, it applies high-capacity convolutional neural networks to bottom-up region recommendations in order to localize and segment objects; and second, it enables more effective training when labeled training data is insufficient.

B. Auto-FPN

The Neural Architecture Search (NAS) technique has demonstrated considerable promise for developing CNNs for image classification applications. Until Auto-FPN, no task-specific or data-oriented network search had been created, and thus no promise of optimal adaptation to any particular dataset could be made. As a result of this, Xu et al. introduced Auto-FPN. Two modules will be included in Auto-FPN: Auto-fusion and Auto-head. Auto-fusion will look for improved multi-level feature fusion, while Auto-head will look for improved classification structure; both modules are also adaptable to a wide variety of popular detector designs and enable fast gradient-based search. We created a completely connected search space with various dilation convolution operations between feature layers in the Auto-fusion module to enable adjustable feature fusion with diverse sensory fields. In the Auto-head module, a more advanced split-transform-merge paradigm should be investigated in order to construct a more efficient and accurate detector rather than the R-CNN head.

Figure 3 illustrates the structure of Auto-search FPN's framework. Our solution is stackable on any backbone and then focuses on the neck and RCNN head of the most appropriate detection system. The Auto-fusion module is designed to improve prediction accuracy; it will seek out a more efficient architecture for utilizing data from all feature levels and performing feature fusion. The Auto-head module is intended to contain a variety of components. The Auto-head module is composed of multiple autonomous search units that provide improved categorization and bounding box prediction.

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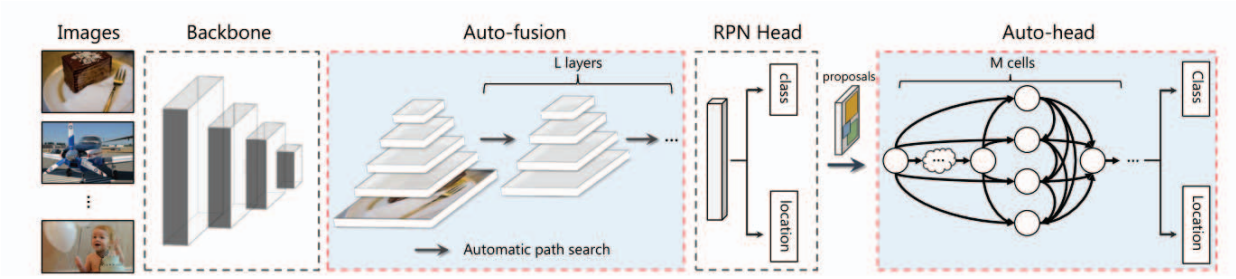

Fig.3 Overall framework of Auto-FPN.

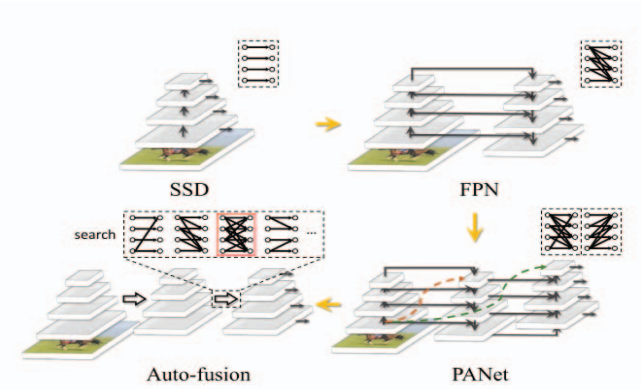

Fig.4 Difference of Auto fusion with other different methods.

The structure of Auto-fusion is depicted in Figure 4. SSD considers a feature pyramid structure directly while utilizing the feature hierarchy for detection. A top-down method proposed with FPN is employed to conduct feature fusion. A bottom-up path enhancement utilizing PANet is presented to further enhance the complete feature hierarchy. After merging the three structures described above, Auto-fusion attempts to determine the ideal neck shape for any detection dataset and task. Auto-fusion processes are capable of covering all three of the aforementioned connection patterns and will attempt to discover a better suitable fusion process via autonomous neural structure search.

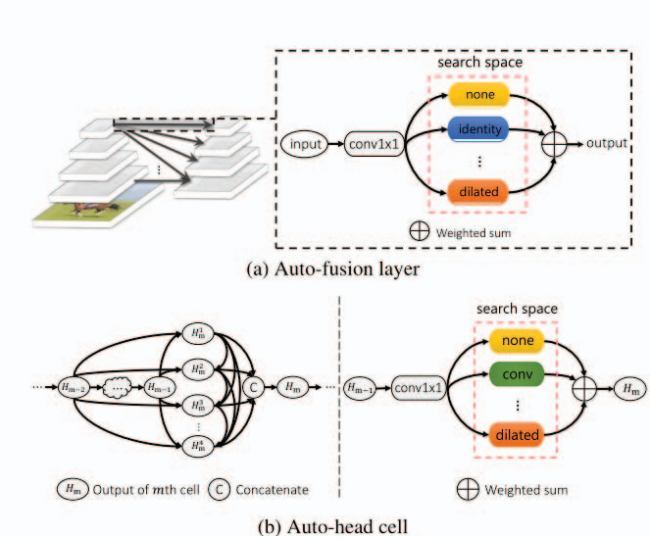

Fig.5 Structure of Auto-head module.

The structure of the Auto-head module is depicted in Figure 5. Seven nodes comprise the Auto-head module: two input nodes, four intermediate nodes, and one output node. All feature levels were connected in their entirety via links consisting of a conv1x1 followed by a collection of parallel operators.

Xu et al. trained with PASCAL VOC, BDD, and COCO. Auto-FPN outperformed the baseline FPN on all three detection benchmarks with less parameters. Table II summarizes the results of Xu et al testing .'s on Auto-FPN. Auto-FPN scored an overall AP of 40.5 percent on COCO, compared to 38.6 percent for FPN; 81.8 percent on VOC, compared to 79.7 percent for FPN; and 33.9 percent on ADE, compared to 33.0 percent for FPN. The searched architectures are around 40%–50% smaller in the neck and 50%–60% smaller in the skull in terms of parameter size. Additionally, it has been discovered that directly searching the target dataset is preferable to transferring the searched structures from other datasets.

TABLE II THE RESULTS OF XU ET AL TESTING .'S ON AUTO-FPN

Method	Search on	mAP
Auto-FPN	COCO	40.5%
Auto-FPN	VOC	81.8%
Auto-FPN	BDD	33.9%
FPN	COCO	38.6%
FPN	VOC	79.7%
FPN	BDD	33%

C. SSPNet

In recent years, due to the emergence of CNN and a large amount of training data, image recognition based on deep learning has developed rapidly, and the fields of image classification and object detection have also been well developed. However, CNN still has technical problems. The size of the input image is fixed, in order to meet this requirement, the image may need to be cropped and distorted, but doing so may cause the content to be lost or distorted and affect the recognition accuracy. In recent years, due to the emergence of CNN and a large amount of training data, image recognition based on deep learning has developed rapidly, and the fields of image classification and object detection have also been well developed. However, CNN still has technical problems. The size of the input image is Fixed, in order to meet this requirement, the image may need to be cropped and distorted, but doing so may cause the content to be lost or distorted and affect the recognition accuracy. He et al. employed another network pooling approach, dubbed "spatial pyramid pooling," to circumvent the aforementioned constraint. The novel network structure, dubbed SPPNet, is capable of generating representations with a fixed length regardless of the image size/scale..

He et al. modified the SSP-net structure by including an SPP layer in the final convolutional layer. The SPP layer gathers features and produces outputs with a defined length that are subsequently fed into fully linked layers.. The SPP layer will read information between the convolutional layer and the fully connected layer to avoid the need for warping

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and clipping at the beginning.

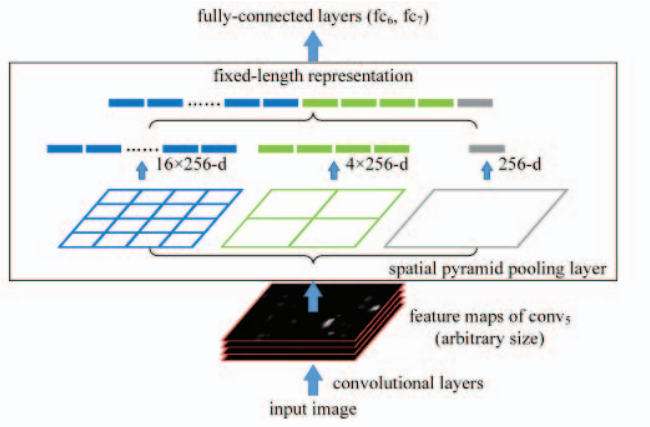

Fig.6 Framework of SPPNet.

The figure 6 above illustrates our approach, where we replace a pooling layer with an SSP layer. Within each spatial bin, we pool the responses of each filter. The output of spatial pyramid pooling is a k-dimensional vector, and the number of bins is denoted as M (k is the number of filters in the last convolutional layer). A vector of fixed dimension is the input to the fully connected layer.

He et al. trained and evaluated the results of SSP-net using the Pascal VOC dataset and compared them with the results of R-CNN. The above figure shows our results on various layers, by using 1-scale or 5-scale. Using pool5 layer, the result is slightly better than that of R-CNN (44.2%). If a non-fine-tuned fc6 layer is used, our results are inferior. But if a fine-tuned layer is used, our results are comparable to or slightly better than R-CNN

The accuracy of SSP-net and R-CNN are basically comparable, but the running time of SSP-net is much shorter than that of R-CNN. This is due to the complexity of SSP-net. Compared with R-CNN, the complexity of 1-scale SSP-net is 1/160. If it is a 5-scale SSP-net, it is only 1/24 of the complexity of R-CNNt

IV. Conclusion

In this paper, a brief review for object detection is conduct, which summarizes five works that have achieved high success in the field of object detection, each of them has its own strengths and weaknesses so may be adapted to different application scenarios in the future. These works may be useful in the application of 3D point clouds, where 2D image to 3D conversion in advanced methods would require massive GPU computation, and tiny networks may be a solution. Our survey will give a prospective guidance of object detection, such as light-weight detector design and object detection by point cloud.

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